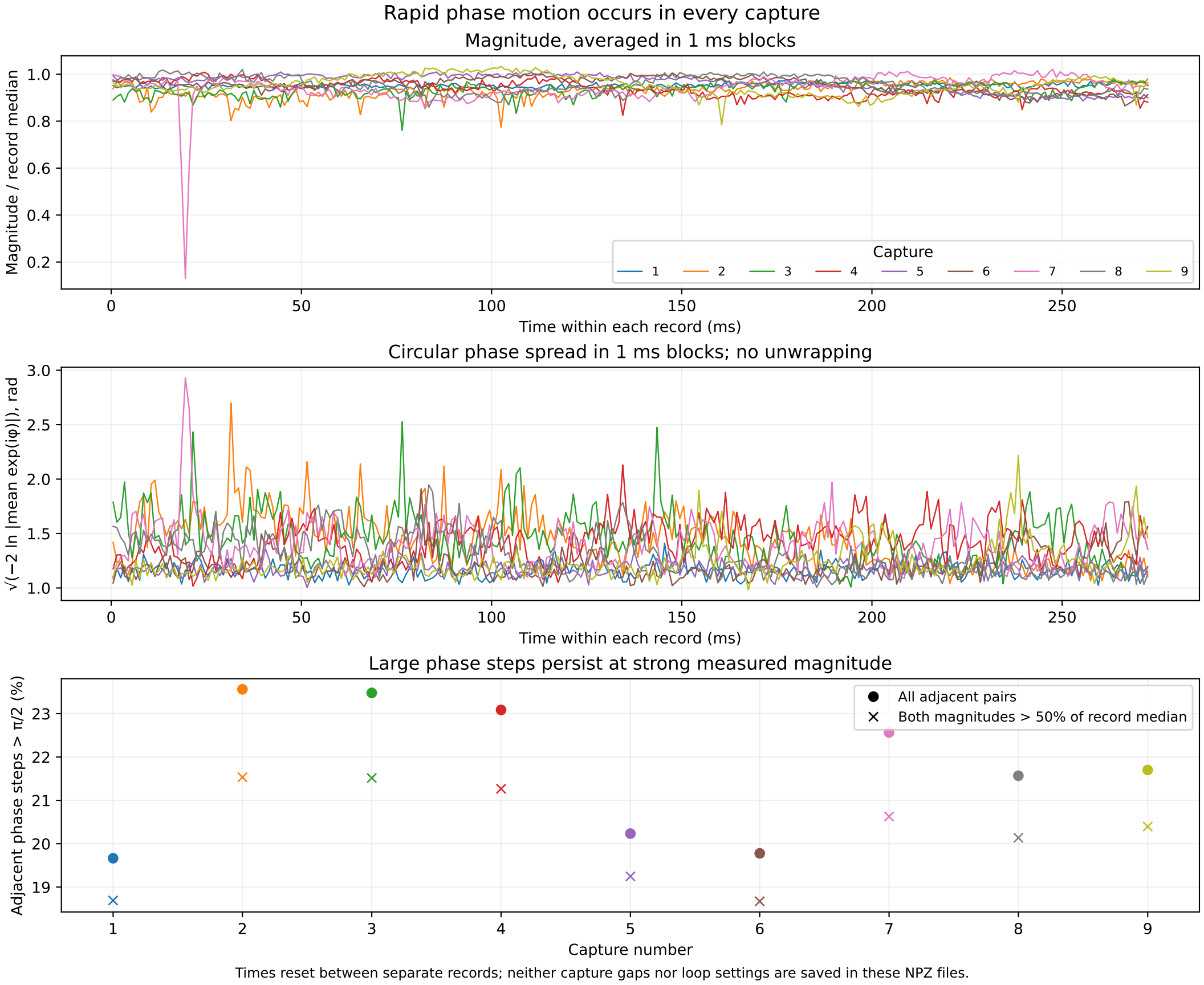

Rapid phase motion occurs in every capture
Magnitude, averaged in 1 ms blocks
Magnitude / record median
1.0
0.8
0.6
0.4
0.2
Capture
1
2
3
4
5
6
7
8
9
0
50
100
150
200
250
Time within each record (ms)
Circular phase spread in 1 ms blocks; no unwrapping
√(−2 ln |mean exp(iφ)|), rad
3.0
2.5
2.0
1.5
1.0
0
50
100
150
200
250
Time within each record (ms)
Large phase steps persist at strong measured magnitude
Adjacent phase steps > π/2 (%)
23
22
21
20
19
All adjacent pairs
Both magnitudes > 50% of record median
1
2
3
4
5
6
7
8
9
Capture number
Times reset between separate records; neither capture gaps nor loop settings are saved in these NPZ files.